Input
TP4 Filter
U3B TL072CDT
These load networks form an LPF with the pickup inductance.
Reducing the guitar volume lowers the Q factor, giving a gentler high-frequency roll-off.
J5
NMJ6HCD3
R20 1k
C13 1u
R21 1M
C14 0.01u
R22 2k2
R23 22k
R27 10k
C16 2200p
C17 2200p
R28 33k
C19 2200p
R32 56k
Q4 2SK880-GR
Q5 2SK880-GR
Q6 2SK880-GR
Control
R24 100M
D16 1N4148WS
R29 100M
D18 1N4148WS
R33 100M
D20 1N4148WS
R41 100M
D22 1N4148WS
D15 1N4148WS
C15 0.1u
D17 1N4148WS
C18 0.1u
D19 1N4148WS
C20 0.1u
D23 1N4148WS
C26 0.1u
R25 220k
R26 1M
R30 220k
R31 1M
R34 220k
R35 1M
R42 220k
R43 1M
Single
Hum
Wah
Effect
To minimize pop, the JFET ahead of the high-gain fuzz stage (Q9) turns on before, and turns off after, the effect output switch Q17. The gate RC cells are asymmetric on purpose; do not make them symmetric.
Q4, Q5, Q6, Q9:
turn-on (1Meg||220k)*0.1u = 18 ms
turn-off 1Meg*0.1u = 100 ms
Q17 (asymmetric the other way):
turn-on (220k+470k)*0.1u = 69 ms
turn-off (690k||470k)*0.1u = 28 ms
Symmetric: Q8 1Meg*0.047u = 47 ms,
Q10, Q11, Q14, Q15 1Meg*0.1u = 100 ms
C25 10u
R40 100k
TP6 FuzzIn
Q9 2SK880-GR
R44 56k
C27 10u
Q12/Q13 hFE: the DC operating point (I_C13 = 300 uA, I_C12 = 30 uA) is set by the R58 feedback. In simulation, hFE values of 340-430 change I_C13 by less than 0.5% and the gain by less than 0.6 dB, mostly via Q13.
R45 6k8
R46 1k5
C28 0.047u
R49 10k
R50 1k
Q10 2SK880-GR
Q11 2SK880-GR
R53 1k
R54 330k
Q12 2SC945-K
I_C13 = 300uA(typ.)
Q13 2SC945-K
TP10 C13
TP9 E13
R58 22k
C35 100p
C34 10u
R56 3k3
R57 3k3
TP8 GND
MP RV1 10kC Fuzz(B1)
R71 1k
C38 10u
R72 1M
R73 47k
FuzzOut
This network interacts with the input impedance of the fuzz stage and attenuates the input signal.
It affects the amount of guitar volume reduction required to prevent saturation.
C35 (100p across R58) adds high-frequency feedback as a precaution. The high-frequency roll-off is set by C33. If you omit it, check for oscillation at all Fuzz and Bias settings.
Control
BIAS30
R70 1M
C37 0.1u
D27 1N4148WS
R69 100M
Q15 2SK880-GR
BIAS15
R65 1M
C36 0.1u
D26 1N4148WS
R64 100M
Q14 2SK880-GR
VCC
Bias adds resistance in parallel with Q13's collector load via Q14/Q15.
(Toggle-Switch Pos.) = [Q14, Q15, Rc(eff), Headroom ratio]
Center = [OFF, OFF, 8.6k, 1:1]
Offset = [ON, OFF, 5.6k, 2:1]
Skew = [OFF, ON, 4.3k, 3:1]
X-Skew = [ON, ON, 3.4k, 4:1]
(Headroom ratio = V(C13)-V(E13) : VCC-V(C13))
J6 A1001WR-S-04PD01
MP
C31 0.1u
C32 0.1u
R55 150k
8.592k
R59 4k7
R60 4k7
R61 100k
16.0k
R62 1k
R63 15k
8.50k
R66 15k
R68 15k
R67 1k
TP7 C12
C33 47p
TP11 C14
R79 2k2
R77 2k7
R78 390
Treble(B2)
C39 0.01u
C40 0.01u
MP RV3 25kB
R83 10k
C44 220p
C45 10u
Pre-Volume
R84 3k3
MP RV4 50kA Volume(A1)
C46 33p
MP RV2 25kB Bass(A2)
R75 2k7
R76 390
U4A OPA1678IDR
U4B OPA1678IDR
C47 1u
R85 47k
Q17 2SK880-GR
C49 1u
R90 15k
R91 220k
pop
Q16 2SK880-GR
R80 2k2
R82 470
C41 0.047u
R81 220k
U3A TL072CDT
v1.1: RV2 and RV3 changed from 20kB to 25kB because the 20kB part could not be sourced. R75/R77 (1k -> 2k7) and R76/R78 (1k5 -> 390) changed with them to keep the v1.0 tone response.
Using low-leakage capacitors helps reduce switching pop noise.
Film capacitors (e.g. PMLCAP) are preferred for C23, C47, and C49 over aluminium electrolytic capacitors.
The fuzz stage output swings up to about 5 Vpp (limited by the 5.6 V VCC). The tone stage adds up to 5x (+14 dB) of treble boost, so peaks can reach the +-8 V rails of U4. Use a rail-to-rail output op-amp.
v1.1: U4 TLV9362IDR -> OPA1678IDR.
TP5 Bypass
Q7 2SJ913-J3
C23 1u
R36 3k3
R37 47k
Q8 2SK880-GR
R38 100M
D21 1N4148WS
R96 100M
B-OUT
R39 1M
C24 0.047u
R37 and R85 must be the same value. The DC step between bypass and effect is dV = Ig * [(R85||R90) - (R37||R90)], so what matters is that they match, not the absolute value. Do not change one without the other. 1% parts are enough: a 2% mismatch adds under 5 uV.
R90: see note near R36.
Q18 2SJ913-J3
C50 10u
R94 1k
R93 100k
R92 10k
J7 NMJ6HCD3
R86 100M
D28 1N4148WS
R97 100M
D29 1N4148WS
R88 470k
C48 0.1u
R87 470k
R89 220k
Effect
H2 North-East
H1 South-West
R90 = 15k and R36 = 3.3k were chosen together. R90 sets the pop dip through Ig * (Rd || Rs). Lowering R90 220k -> 15k with matched R37/R85 (100k/220k -> 47k/47k) reduces pop.
Raising R90 gives some back.
R36 = 3.3k restores the negative swing lost to the low R90.
At 10k, R36 costs bypass THD and clipping headroom. For more low end, change C49 to 2.2u instead of raising these resistors.
R38: see Rg note below R86.
Rg (R38 + R96, R86 + R97) = 200M discharges the gate charge when the JFET switch turns on.
The series diode only handles turn-off; it cannot do this.
Do not omit Rg: recovery would then depend on diode leakage alone.
Lowering it increases pop.
v1.1: Rg changed from 100M to 200M (two 100M in series).
Control
SW1 2MS3-T1-B4-M2-Q-E Input-Filter(C3)
SW2 2MS3-T1-B4-M2-Q-E Response(C2)
SW3 2MS3-T1-B4-M2-Q-E Bias(C1)
J4 FH-2x10SG
Effect
B-OUT
BIAS15
BIAS30
Single
Hum
Wah
Medium
Gradual
South-East
R19 3k3
D14 LED(RED) p5V
Note the PL9823-F5 pinout. In the datasheet package drawing (leads down), the leads run 4(DIN), 3(VDD), 1(GND), 2(DO) from left to right.
The symbol and the standard 4-pin THT LED footprint number them 4(DIN), 3(VDD), 2(GND), 1(DO): pins 1 and 2 are swapped in number only, and the physical order is unchanged.
TP2 DIN
TP3 DO
D13 LED_PL9823-F5
R17 33
R18 1k
C12 0.1u
TIM
Control
Medium
Gradual
R48 1M
C29 0.1u
D24 1N4148WS
R47 100M
R52 1M
C30 0.1u
D25 1N4148WS
R51 100M
power
v1.1: D1 MMSZ5245B -> BAS416.
MJ-179PH
J1
J9
MP
S2B-ZR-SM4A-TF(LF)(SN)
v1.1: J9 added for remote sensing of the supply voltage.
Q1 SSM3J334R
Q2 SS8550 Y2
Q3 SSM3J334R
D1 BAS416
D2 MMSZ5245B
R1 100k
R2 33k
R3 10k
D3 BZT52C13
D30 1N4148WS
C51 1u
R95 1M
R4 470k
C1 1u
R5 47k
C2 470u
25PZJ470M10X9
R6 6k8
D4 LED(RED) p9V
v1.1: D30, C51 and R95 added for soft start (limits inrush current).
v1.1: D3 BZT52C15 -> BZT52C13.
MIC5205 defines Vref as Vout - Vadj, not as the usual bottom-leg reference. R7 22k sits between ADJ and GND; R8 + R9 = 6.2k sits between OUT and ADJ. Vout = 5.65 V.
Applying the ordinary divider formula to this gives 1.59 V, which is wrong.
D5 BAT54S
U1 MIC5205YM5
C3 10uF
C4 0.1u
R7 22k
C5 470p
R8 1k5
R9 4k7
R11 22
R10 4.7
C6 4.7u
C7 100u 16SEPC100M
Vout=5.65V
Vout-Vadj=1.242V
Rbtm=22k
Rtop=6.2k
C8 10uF
C9 10uF
D6 1N5819WS
D7 1N5819WS
C10 10uF
D8 1N5819WS
R12 6k8
D9 LED(RED) p8V
R13 3k3
D10 LED(RED) Vcc
R14 6k8
D11 LED(RED) n8V
J8 B6B-PH-K-S
Input
Bypass
FuzzOut
Pre-Volume
pop
J10 PH-1X2SG
J10: for measurement only, not fitted.
Leave open: shorting it ties Bypass to FuzzOut.
J2 MJ-8435
MIDI IN (MIDI TRS Type A compliant)
MIDI-GND
R15 220
D12 1N4148WS
C11 0.1u
R16 4k7
U2 TLP2361
To keep MIDI IN edges sharp, use a photocoupler with a response time of 1 us or less.
USART
J3 FH-1x4SG/RH North-West
+9V
+8V
-8V
+5V

Source: github.com/trope-oshw/anthracite
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Yamamoto Circuit Design
Sheet: /
File: anthracite-top.kicad_sch

Title: Anthracite Schematic (Top)

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